When is a Measurement RG-Relevant?

A step-by-step corrected guide

How to read this note

Every important algebraic step is written out. In particular, whenever we use

$$y_\gamma = d - 2\Delta_{\mathcal{O}},$$

we first state the measured operator $\mathcal{O}$, then state $\Delta_{\mathcal{O}}$, then substitute it line by line. There is no shorthand like "$d - 2(d-2)/2 = 2$".

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1 The replica trick for measurements (done carefully)

1.1 Setup

We have an unperturbed CFT with action $S_*[\phi]$ in d Euclidean dimensions, and we measure a local operator $\phi(x)$ at every point with a Gaussian detector of variance Δ. The likelihood for an outcome record $m(x)$ given the field configuration is

$$P[m|\phi] \;\propto\; \exp\left[-\frac{1}{2\Delta}\int \mathrm{d}^d x\,\big(m(x) - \phi(x)\big)^2\right]. \tag{1}$$

The Born rule weights the outcome by $|\text{wavefunction}|^2$; in the path-integral formulation this means each outcome's probability is controlled by two copies (forward and backward) of the

path integral. For physical observables averaged over Born-distributed outcomes one needs

$$\overline{\langle \mathcal{O} \rangle} = \int \mathcal{D}m \; P[m] \, \langle \mathcal{O} \rangle_m = \lim_{R \to 1} \int \mathcal{D}m \; Z[m]^{R-1} \, \langle \mathcal{O} \rangle_m \, Z[m], \tag{2}$$

where the factor $Z[m]^{R-1}$ raised to a non-integer power is what the replica limit $R \to 1$ handles. Crucially the R copies all couple to the *same* outcome field m.

1.2 Gaussian integration over the outcome

Now do the Gaussian integral over $m(x)$ in the replicated theory. Pointwise (suppressing x), define

$$A \equiv \sum_{a=1}^{R} \phi^{(a)}.$$

Then

$$\begin{aligned}
\sum_{a=1}^{R} (m - \phi^{(a)})^2 &= \sum_{a=1}^{R} \left[m^2 - 2m\phi^{(a)} + \phi^{(a)\,2} \right] \\
&= Rm^2 - 2m \sum_a \phi^{(a)} + \sum_a \phi^{(a)\,2} \\
&= Rm^2 - 2mA + \sum_a \phi^{(a)\,2} \\
&= R \left(m - \frac{A}{R} \right)^2 - \frac{A^2}{R} + \sum_a \phi^{(a)\,2}.
\end{aligned} \tag{3}$$

Therefore

$$\begin{aligned}
&\int \mathrm{d}m \; \exp\left[-\frac{1}{2\Delta} \sum_{a=1}^{R} (m - \phi^{(a)})^2 \right] \\
&\quad = \int \mathrm{d}m \; \exp\left[-\frac{R}{2\Delta} \left(m - \frac{A}{R} \right)^2 + \frac{A^2}{2\Delta R} - \frac{1}{2\Delta} \sum_a \phi^{(a)\,2} \right] \\
&\quad \propto \exp\left[-\frac{1}{2\Delta} \sum_a \phi^{(a)\,2} + \frac{1}{2\Delta R} \Big(\sum_a \phi^{(a)} \Big)^2 \right].
\end{aligned} \tag{4}$$

In the second line, the Gaussian integral over $m - A/R$ only gives an overall constant, so it can be dropped.

Now expand the remaining square:

$$\begin{aligned}
\left(\sum_a \phi^{(a)} \right)^2 &= \sum_a \sum_b \phi^{(a)} \phi^{(b)} \\
&= \sum_a \phi^{(a)\,2} + \sum_{a \neq b} \phi^{(a)} \phi^{(b)} \\
&= \sum_a \phi^{(a)\,2} + 2 \sum_{a<b} \phi^{(a)} \phi^{(b)}.
\end{aligned} \tag{5}$$

Substituting this into Eq. (4) gives

$$
\begin{aligned}
&-\frac{1}{2\Delta}\sum_a \phi^{(a)\,2} + \frac{1}{2\Delta R}\left[\sum_a \phi^{(a)\,2} + 2\sum_{a<b}\phi^{(a)}\phi^{(b)}\right] \\
&= -\frac{R}{2\Delta R}\sum_a \phi^{(a)\,2} + \frac{1}{2\Delta R}\sum_a \phi^{(a)\,2} + \frac{1}{\Delta R}\sum_{a<b}\phi^{(a)}\phi^{(b)} \\
&= -\frac{R-1}{2\Delta R}\sum_a \phi^{(a)\,2} + \frac{1}{\Delta R}\sum_{a<b}\phi^{(a)}\phi^{(b)}. \qquad (6)
\end{aligned}
$$

So the integrated weight is

$$
\exp\left[-\frac{R-1}{2\Delta R}\sum_a \phi^{(a)\,2} + \frac{1}{\Delta R}\sum_{a<b}\phi^{(a)}\phi^{(b)}\right]. \qquad (7)
$$

Reading off, the replicated action gains

$$
\boxed{\delta S = -\frac{1}{\Delta R}\int \mathrm{d}^d x \sum_{a<b}\phi^{(a)}(x)\phi^{(b)}(x) + \frac{R-1}{2\Delta R}\int \mathrm{d}^d x \sum_a \phi^{(a)\,2}(x).} \qquad (8)
$$

Two observations.

- The off-diagonal piece $\sum_{a<b}\phi^{(a)}\phi^{(b)}$ is the central object. Calling its coupling $\gamma \equiv 1/(\Delta R)$, the perturbation has the symmetric form $-\gamma\sum_{a\neq b}\phi^{(a)}\phi^{(b)}/2$.
- The diagonal piece $\sum_a \phi^{(a)\,2}$ has prefactor $R-1$ and *vanishes* when $R \to 1$.[1] For tree-level relevance this term is irrelevant ($R \to 1$ kills it). For a Gaussian theory, however, keeping it makes the replica locking spectrum manifest; see Sec. 3.

The only fact about measurements we need

After Born-averaging and replication, weak Gaussian measurement of $\phi(x)$ in a CFT S_* adds, in the $R \to 1$ limit,

$$
\delta S = -\gamma \int \mathrm{d}^d x \sum_{a\neq b}^{R} \phi^{(a)}(x)\,\phi^{(b)}(x), \qquad \gamma \sim 1/\Delta,
$$

plus single-replica terms that vanish as $R \to 1$. Everything else is standard RG applied to this perturbation; the $R \to 1$ limit distinguishes measurement from quenched disorder ($R \to 0$).

2 Power counting: the tree-level relevance criterion

2.1 Master formula

Under $x \to bx$ a primary operator ϕ of scaling dimension Δ_ϕ transforms as $\phi(x) \to b^{-\Delta_\phi}\phi(x/b)$. The integration measure transforms as

$$
\mathrm{d}^d x \to b^d\,\mathrm{d}^d x.
$$

[1] Strictly: the diagonal piece is multiplied by a single-replica integration measure that survives $R \to 1$, but its coefficient does not.

The product of two measured operators in different replicas transforms as

$$\phi^{(a)}(x)\phi^{(b)}(x) \to b^{-\Delta_\phi} b^{-\Delta_\phi} \phi^{(a)}(x/b)\phi^{(b)}(x/b) = b^{-2\Delta_\phi}\phi^{(a)}(x/b)\phi^{(b)}(x/b).$$

Therefore the full perturbing integral transforms as

$$\begin{aligned}\int \mathrm{d}^d x\, \phi^{(a)}\phi^{(b)} &\to b^d b^{-2\Delta_\phi} \int \mathrm{d}^d x\, \phi^{(a)}\phi^{(b)} \\ &= b^{d-2\Delta_\phi} \int \mathrm{d}^d x\, \phi^{(a)}\phi^{(b)}.\end{aligned} \tag{9}$$

For the perturbation in (8), the coupling must therefore scale as

$$\gamma \to b^{y_\gamma}\gamma.$$

The exponent is

$$\boxed{y_\gamma \;=\; d - 2\Delta_\phi.} \tag{10}$$

Harris–Luck criterion for measurements (tree level)

$$2\Delta_\phi < d \;\Rightarrow\; \text{relevant}; \qquad 2\Delta_\phi = d \;\Rightarrow\; \text{marginal}; \qquad 2\Delta_\phi > d \;\Rightarrow\; \text{irrelevant}.$$

Remark. This is identical in form to the Harris criterion for bond disorder but the replica limit differs ($R \to 1$ vs. $R \to 0$). The relevance/irrelevance question is settled at tree level away from marginality. At marginality, and for the new IR universality class, the $R \to 1$ loop coefficients matter.

3 Toy model 1: free massless scalar, measure ϕ

Setup. The action is

$$S_* = \frac{1}{2}\int \mathrm{d}^d x\, (\partial\phi)^2.$$

At a scale-invariant fixed point the action is dimensionless. The dimensions are

$$[\mathrm{d}^d x] = -d, \qquad [\partial] = 1, \qquad [\phi] = \Delta_\phi.$$

The integrand has dimension

$$[(\partial\phi)^2] = 2[\partial] + 2[\phi] = 2 + 2\Delta_\phi.$$

For the action to be dimensionless,

$$\begin{aligned}[\mathrm{d}^d x] + [(\partial\phi)^2] &= 0, \\ -d + (2 + 2\Delta_\phi) &= 0, \\ 2 + 2\Delta_\phi &= d, \\ 2\Delta_\phi &= d - 2, \\ \Delta_\phi &= \frac{d-2}{2}.\end{aligned} \tag{11}$$

Tree level. Now use the master formula

$$y_\gamma = d - 2\Delta_\phi.$$

Substitute $\Delta_\phi = (d-2)/2$:

$$\begin{aligned} y_\gamma &= d - 2\left(\frac{d-2}{2}\right) \\ &= d - \left[\frac{2(d-2)}{2}\right] \\ &= d - (d-2) \\ &= d - d + 2 \\ &= 2. \end{aligned} \tag{12}$$

Strongly relevant in every dimension.

Why this is also exact. The replicated perturbation is quadratic in ϕ, so the full theory is still Gaussian. If one keeps only the off-diagonal term $-\gamma \sum_{a<b} \phi^{(a)}\phi^{(b)}$, then the quadratic form is

$$\delta S_{\text{off}} = -\gamma \sum_{a<b} \phi_a \phi_b = -\frac{\gamma}{2}\left[\left(\sum_a \phi_a\right)^2 - \sum_a \phi_a^2\right].$$

For the symmetric mode, all replicas are equal:

$$\phi_1 = \phi_2 = \cdots = \phi_R = s.$$

Then

$$\sum_a \phi_a = Rs, \qquad \sum_a \phi_a^2 = Rs^2,$$

so

$$\delta S_{\text{off}} = -\frac{\gamma}{2}(R^2 s^2 - Rs^2) = -\frac{\gamma}{2} R(R-1)s^2.$$

Thus the symmetric direction has negative mass-squared proportional to $-(R-1)\gamma$ in the off-diagonal-only approximation.

For an antisymmetric mode,

$$\sum_a \phi_a = 0.$$

Then

$$\delta S_{\text{off}} = -\frac{\gamma}{2}(0 - \sum_a \phi_a^2) = +\frac{\gamma}{2}\sum_a \phi_a^2,$$

so the antisymmetric modes have positive mass-squared proportional to $+\gamma$.

If one instead keeps the exact Gaussian integration result in Eq. (8), the quadratic perturbation is

$$\begin{aligned} \delta S_{\text{exact}} &= -\frac{1}{\Delta R}\sum_{a<b} \phi_a \phi_b + \frac{R-1}{2\Delta R}\sum_a \phi_a^2 \\ &= -\frac{1}{2\Delta R}\left[\left(\sum_a \phi_a\right)^2 - \sum_a \phi_a^2\right] + \frac{R-1}{2\Delta R}\sum_a \phi_a^2 \\ &= -\frac{1}{2\Delta R}\left(\sum_a \phi_a\right)^2 + \frac{1}{2\Delta}\sum_a \phi_a^2. \end{aligned} \tag{13}$$

For the symmetric mode $\phi_a = s$ this gives

$$\delta S_{\text{exact}} = -\frac{1}{2\Delta R}(Rs)^2 + \frac{1}{2\Delta}(Rs^2) = -\frac{R}{2\Delta}s^2 + \frac{R}{2\Delta}s^2 = 0.$$

For an antisymmetric mode $\sum_a \phi_a = 0$, it gives

$$\delta S_{\text{exact}} = 0 + \frac{1}{2\Delta}\sum_a \phi_a^2.$$

Thus the symmetric physical mode is untouched, while the relative replica modes get mass-squared $1/\Delta$. The RG eigenvalue $y_\gamma = 2$ is just $[m^2] = 2$ and is not corrected at any order.

4 Toy model 2: measure ϕ^2 at the Wilson–Fisher fixed point in $d = 4 - \epsilon$

This is the example v1 of this note got wrong; redoing it correctly is the most important exercise in the document.

4.1 The mistake to avoid

In free ϕ^4 theory at $d = 4$, ϕ^2 has classical dimension $d - 2 = 2$. It is tempting (and wrong) to declare $\Delta_{\phi^2} = d - 2$ to leading order in ϵ. This ignores the $O(\epsilon)$ anomalous dimension that ϕ^2 picks up at Wilson–Fisher, and the resulting Harris eigenvalue is wrong by a factor of 3.

4.2 The correct Δ_ε at Wilson–Fisher

At the Ising universality class fixed point in $d = 4 - \epsilon$ the energy operator $\varepsilon \sim \phi^2$ has dimension fixed by the correlation-length exponent via the hyperscaling/Josephson relation:

$$\Delta_\varepsilon = d - \frac{1}{\nu}. \tag{14}$$

At one loop,

$$\nu = \frac{1}{2} + \frac{\epsilon}{12} + O(\epsilon^2). \tag{15}$$

Now invert it explicitly:

$$\begin{aligned}
\nu &= \frac{1}{2}\left(1 + \frac{\epsilon}{6}\right) + O(\epsilon^2), \\
\frac{1}{\nu} &= \frac{2}{1 + \epsilon/6} + O(\epsilon^2), \\
&= 2\left(1 - \frac{\epsilon}{6}\right) + O(\epsilon^2), \\
&= 2 - \frac{\epsilon}{3} + O(\epsilon^2).
\end{aligned} \tag{16}$$

Therefore

$$\begin{aligned}
\Delta_\varepsilon &= d - \frac{1}{\nu} \\
&= (4 - \epsilon) - \left(2 - \frac{\epsilon}{3}\right) + O(\epsilon^2) \\
&= 4 - \epsilon - 2 + \frac{\epsilon}{3} + O(\epsilon^2) \\
&= 2 - \frac{2\epsilon}{3} + O(\epsilon^2).
\end{aligned} \tag{17}$$

The anomalous part is $-2\epsilon/3$, not 0.

4.3 The Harris eigenvalue

$$
\begin{aligned}
y_\gamma &= d - 2\Delta_\varepsilon \\
&= (4-\epsilon) - 2\left(2 - \frac{2\epsilon}{3}\right) + O(\epsilon^2) \\
&= 4 - \epsilon - \left(4 - \frac{4\epsilon}{3}\right) + O(\epsilon^2) \\
&= 4 - \epsilon - 4 + \frac{4\epsilon}{3} + O(\epsilon^2) \\
&= -\epsilon + \frac{4\epsilon}{3} + O(\epsilon^2) \\
&= \boxed{\frac{\epsilon}{3} + O(\epsilon^2).}
\end{aligned}
\tag{18}
$$

This is exactly Harris's relation $y_\gamma = \alpha/\nu$ in disguise: the specific-heat exponent $\alpha = 2 - d\nu = \epsilon/6 + O(\epsilon^2)$, and $\alpha/\nu = \epsilon/3 + O(\epsilon^2)$.

Conclusion. Weak measurement of the energy density at the Wilson–Fisher fixed point is relevant for $\epsilon > 0$ (*i.e.* $d < 4$), with a Harris-like eigenvalue $y_\gamma = \epsilon/3$, marginal at $d = 4$, irrelevant for $d > 4$. The smallness of y_γ in ϵ is what makes the perturbative RG controlled near the upper critical dimension.

Remark (3D Ising)**.** Beyond the ϵ-expansion, the numerically known 3D Ising dimension $\Delta_\varepsilon \approx 1.413$ gives $y_\gamma = 3 - 2(1.413) \approx 0.175 > 0$: energy measurement is (weakly) relevant in 3D Ising, consistent with our $\epsilon/3$ result extrapolated to $\epsilon = 1$ ($\epsilon/3 \approx 0.33$, in the right direction).

5 Toy model 3: measure σ and ε in 2D critical Ising

The 2D Ising CFT has exact primary dimensions

$$
\Delta_\sigma = \frac{1}{8}, \qquad \Delta_\varepsilon = 1, \tag{19}
$$

and the famous fusion rules

$$
\sigma \times \sigma = \mathbb{1} + \varepsilon, \qquad \sigma \times \varepsilon = \sigma, \qquad \varepsilon \times \varepsilon = \mathbb{1}. \tag{20}
$$

5.1 Measuring the spin

For spin measurement the measured operator is

$$
\mathcal{O} = \sigma, \qquad \Delta_\mathcal{O} = \Delta_\sigma = \frac{1}{8}.
$$

Therefore

$$
\begin{aligned}
y_\gamma &= d - 2\Delta_\sigma \\
&= 2 - 2\left(\frac{1}{8}\right) \\
&= 2 - \frac{2}{8} \\
&= 2 - \frac{1}{4} \\
&= \frac{8}{4} - \frac{1}{4} \\
&= \frac{7}{4}.
\end{aligned}
\tag{21}
$$

Strongly relevant. The IR is governed by a new fixed point related to the Nishimori line of the random-bond Ising model in its measurement incarnation (see Wiese–Nahum, arXiv:2604.23346, for the exact treatment using enlarged replica symmetry).

5.2 Measuring the energy

For energy measurement the measured operator is

$$\mathcal{O} = \varepsilon, \qquad \Delta_{\mathcal{O}} = \Delta_\varepsilon = 1.$$

Therefore

$$\begin{aligned} y_\gamma &= d - 2\Delta_\varepsilon \\ &= 2 - 2(1) \\ &= 2 - 2 \\ &= 0. \end{aligned} \tag{22}$$

So energy measurement is tree-level marginal. The perturbation is

$$\delta S = -\gamma \int \mathrm{d}^2 x \sum_{a<b} \varepsilon_a(x)\varepsilon_b(x). \tag{23}$$

Whether it is marginally relevant or marginally irrelevant requires the one-loop β-function.

The subtle point. It is true that $C_{\varepsilon\varepsilon\varepsilon} = 0$ in the single-copy 2D Ising CFT:

$$\varepsilon(x)\,\varepsilon(0) \sim \frac{1}{|x|^2}\,\mathbb{1} + \cdots, \qquad \varepsilon \times \varepsilon = \mathbb{1}. \tag{24}$$

v1 correctly needed to avoid attributing this to $\mathbb{Z}_2$ symmetry: ε is $\mathbb{Z}_2$-even, so spin-flip symmetry does not forbid ε^3. The reason is the minimal-model fusion rule, equivalently the Virasoro/Kramers–Wannier structure.

However, $C_{\varepsilon\varepsilon\varepsilon} = 0$ does *not* make the replicated one-loop beta function vanish. The product of two replica bilinears can share only one replica index, for example

$$\big[\varepsilon_a\varepsilon_b\big](x)\big[\varepsilon_a\varepsilon_c\big](0) \sim \frac{1}{|x|^2}\,\varepsilon_b(0)\varepsilon_c(0), \qquad b \neq c. \tag{25}$$

This regenerates the same off-diagonal perturbation. Counting the allowed shared-index contractions gives the standard replicated Potts/Ising beta function

$$\frac{\mathrm{d}\gamma}{\mathrm{d}\ell} = y_\gamma\gamma + 4\pi(R-2)\gamma^2 - 16\pi^2(R-2)\gamma^3 + \cdots. \tag{26}$$

For the Ising energy operator, $y_\gamma = 0$. In the measurement replica limit $R \to 1$,

$$\begin{aligned} \frac{\mathrm{d}\gamma}{\mathrm{d}\ell} &= y_\gamma\gamma + 4\pi(R-2)\gamma^2 - 16\pi^2(R-2)\gamma^3 + \cdots \\ &= 0\cdot\gamma + 4\pi(1-2)\gamma^2 - 16\pi^2(1-2)\gamma^3 + \cdots \\ &= 0 + 4\pi(-1)\gamma^2 - 16\pi^2(-1)\gamma^3 + \cdots \\ &= -4\pi\gamma^2 + 16\pi^2\gamma^3 + \cdots. \end{aligned} \tag{27}$$

Therefore weak positive energy measurement in the 2D critical Ising model is *marginally irrelevant*: for small positive γ, the leading term is $-4\pi\gamma^2 < 0$, so γ decreases under RG. It flows back to the unmeasured Ising fixed point at sufficiently small measurement strength, although a separate finite-strength transition can still occur.

Further reading

- A. Nahum, K. J. Wiese, *Renormalization group for measurement and entanglement phase transitions*, arXiv:2303.07848.
- A. Nahum, J. Jacobsen, *Bayesian critical points in classical lattice models*, arXiv:2504.01264.
- K. J. Wiese, A. Das, A. Nahum, *Bayesian phase transition for the critical Ising model: enlarged replica symmetry in the epsilon expansion and in 2D*, arXiv:2604.23346.
- R. A. Patil, A. W. W. Ludwig, *A c-theorem for the effective central charge in the $R = 1$ replica limit, and applications to systems with measurement-induced randomness.*